From the Geo-INQUIRE SDL to our NFDI4Earth LUC: RDM for SeisSol

2026-10-12

NFDI4Earth IG HPC (zoom)

iris.christadler@lmu.de, fynn.graf@campus.lmu.de, mathilde.marchandon@lmu.de, alice.gabriel@ucsd.edu, heiner.igel@lmu.de, bader@cit.tum.de

Geo-INQUIRE *is funded by the European Union. Views and opinions expressed are however those of the authors only and do not necessarily reflect those of the European Union or the European Research Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.*

Agenda

- Simulation Data Lake@CINECA
- SeisSol & European Projects
- SeisSol_ParamStudies (Ad 1/2)
- NFDI4Earth LUC:
  RDM Workflow & AI Readiness
- Geo-INQUIRE
  Transnational Access Datasets
- SDL_Metadata_Generator (Ad 2/2)

SDL:

- SDL: https://sdl.hpc.cineca.it
- SDL Documentation:
  - https://geo-inquire.pages.hpc.cineca.it/sdl/tutorials/Introduction/ (webinterface)
  - https://geo-inquire.pages.hpc.cineca.it/sdlctl/ (sdlctl, pythonSDK, reference)

SeisSol Research Data Management Workflow for LUC:

- SeisSol_ParamStudies:
  https://github.com/christadler/SeisSol_ParamStudies
- SDL_Metadata_Generator
  https://github.com/fynngraf/SDL_Metadata_Generator
- SDL_Tools:
  https://github.com/christadler/SDL_Tools

Other DT-Geo Workflow Components:

- SeisSol_Postprocessing (and SDL upload):
  https://github.com/christadler/SeisSol_Postprocessing
- Query TABOO NFO and cronjob setup:
  https://github.com/marcmath/download_TABOO_waveforms
- Find closest match from catalog:
  https://github.com/NicoSchlw/search_SeisSol_ensemble

Simulation Data Lake@CINECA

The Geo-INQUIRE Simulation Data Lake

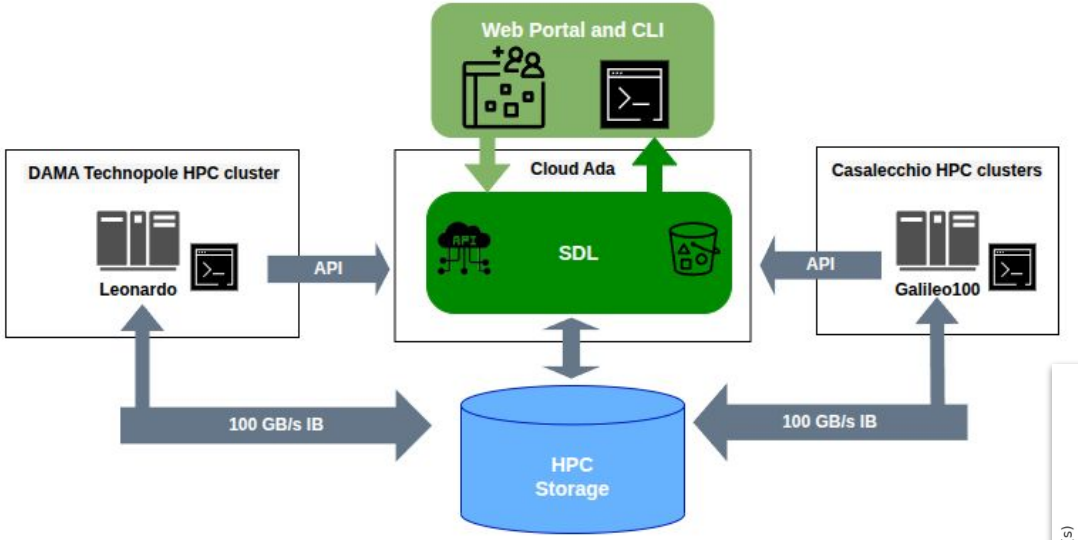
Access to SDL:
https://sdl.hpc.cineca.it

- **Data portal for simulated data** (inputs, results & data products)
- **Easy access:**
  - Get account and store data,
  - Webinterface, CLI, python package and API
  - Webinterface makes it easy to search or up-/download data
- **Fast access:** Upload GB in minutes
- **Metadata**:
  - Per dataset (file), simulation run (folder), experiment
  - Complete set (incl. DOI), all datacite metadata can be added,
  - Metadata defined to extend it for certain datasets
- **Long-term storage**

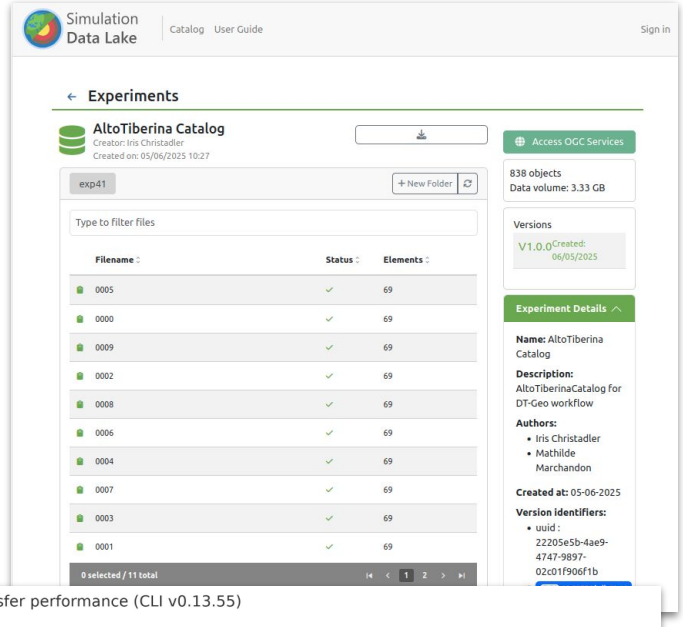
Zenodo File Size limitations:
50 GB per record, max. 100 files
one-time quote of 200GB possible (upon request), use ZIP if you have more files (already for 20+ files)
Simulation Data Lake:
Dev: in total 1 TB, Prod: 500 TB in total

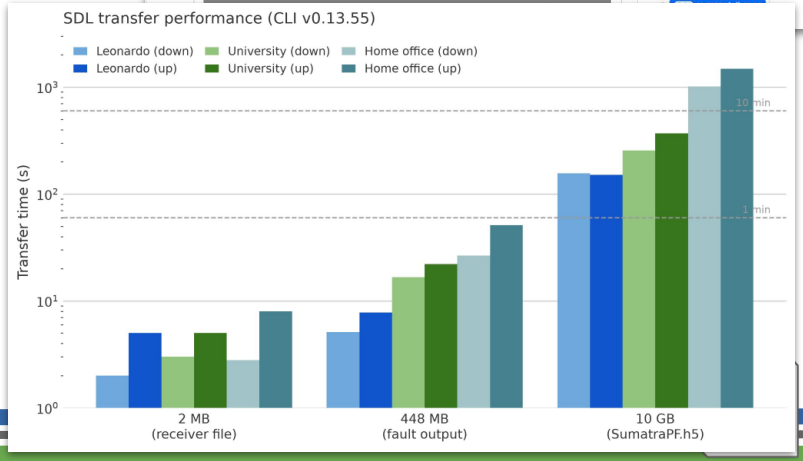
… a **Research Data Management (RDM) solution for SeisSol**:
defined workflows to save SeisSol input data, meshes and outputs in a uniform way to ensure AI-readiness

Simulation Data Lake SDL@CINECA



©CINECA

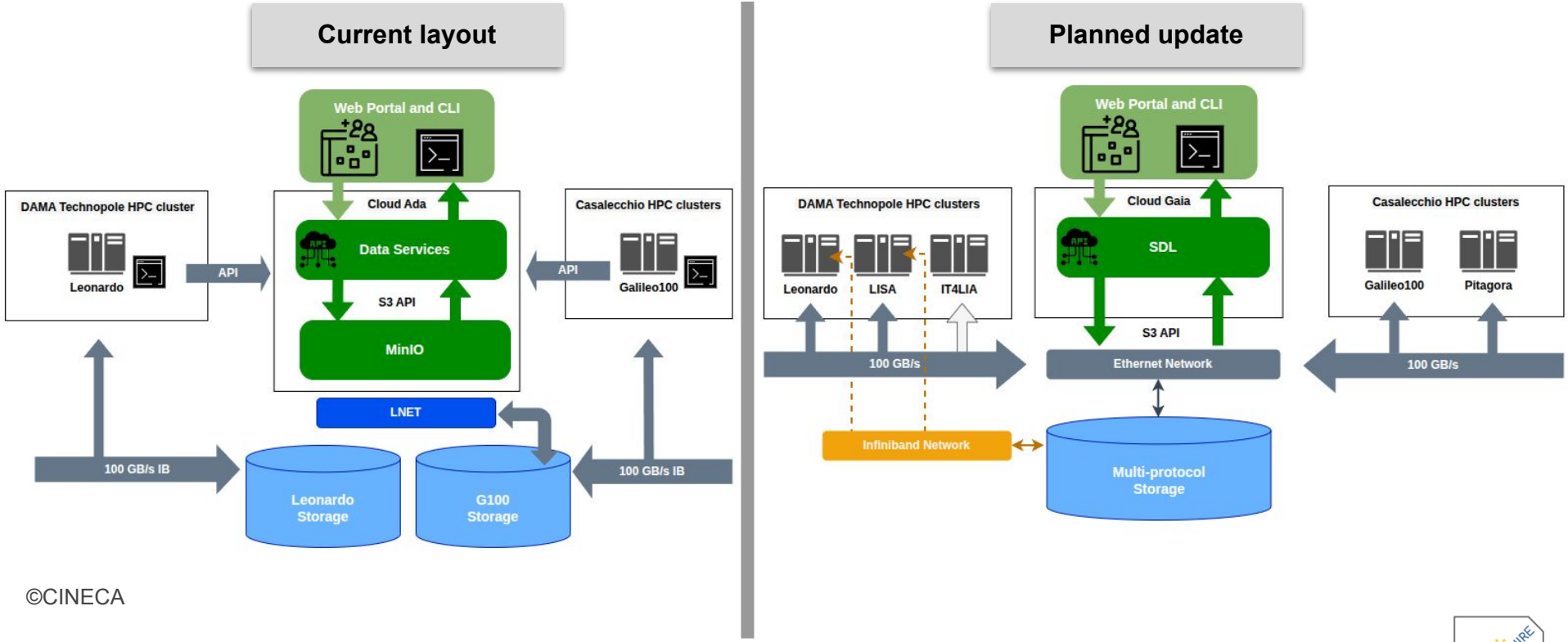




Access to SDL:
https://sdl.hpc.cineca.it

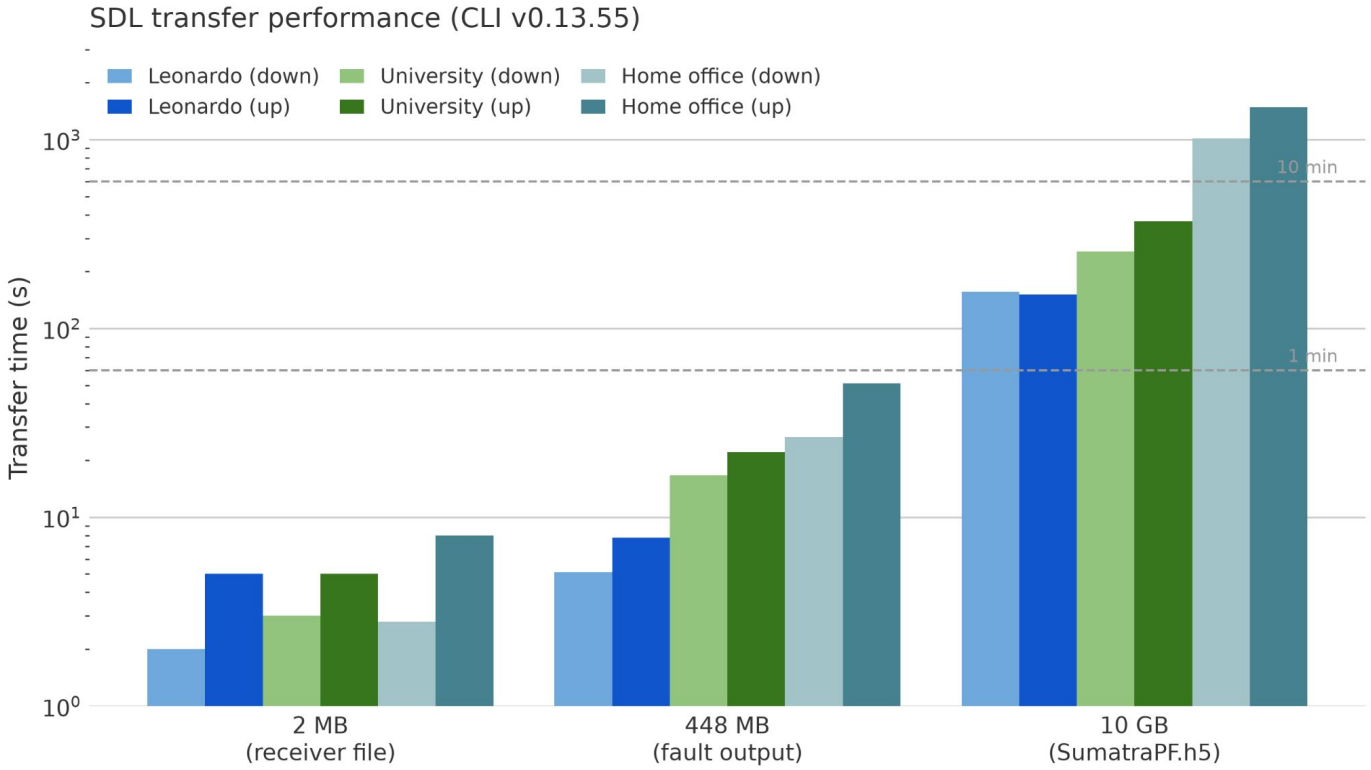
SDL is still on Ada Cloud
and uses MinIO
(not yet on Gaia with VAST)
“If POSIX access is made available,
it will most likely be read-only”

Simulation Data Lake SDL@CINECA



©CINECA

Benchmarking SeisSol Datasets on the SDL: Transfer time



SDL Metadata Scheme

Experiment

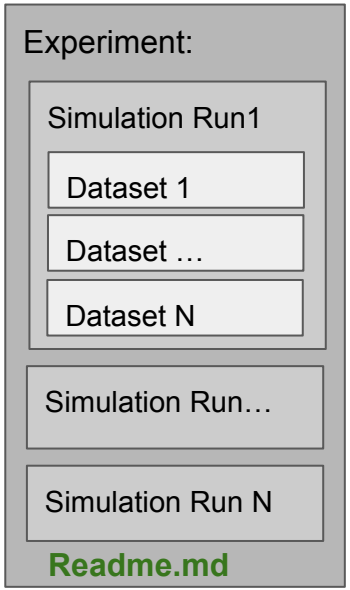
- Name+Description
- Author(s)
- Organization(s)
- Project(s)
- Collaborator(s) (Editor|Viewer)
- Keywords&DOI
- *a way to add any datacite object…*

Simulation Run

- Name+Description
- Path

Workflow Files

- Common Workflow Language (CWL) file



Dataset:

- File Type (input|output|data product|input parameter)
- Spatial Coverage (Geographical location)
- Start+End Date (Temporal location)
- File Format (TXT, CSV, HDF5, XDMF, …)
- Description

Readme.(txt|md):

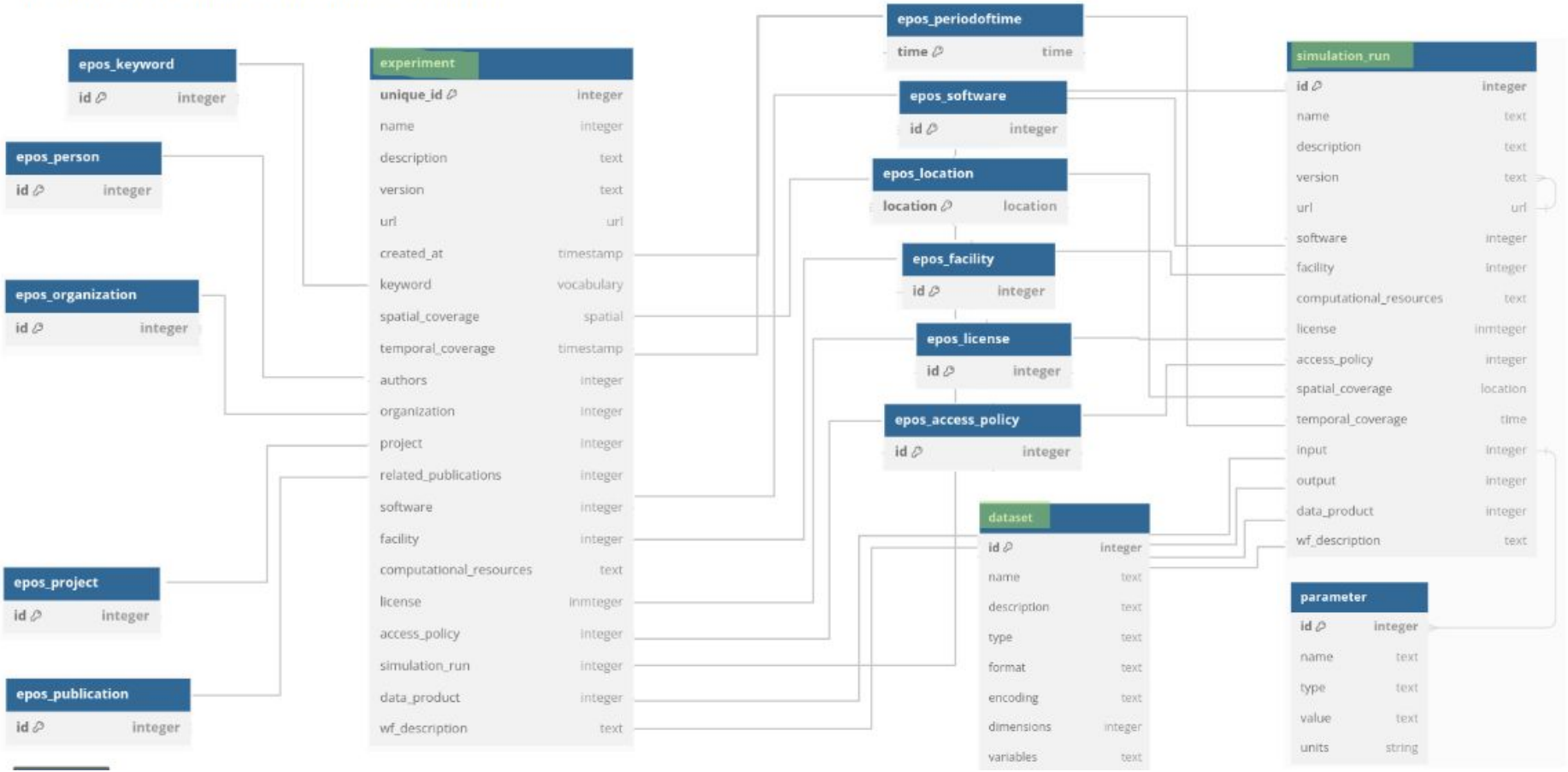
- Will be displayed on first page
- Used by LMU's SDL_Metadata_Generator

Superficial view, the actual SDL metadata scheme is aligned with EPOS-DCAT_AP to allow integration of datasets into EPOS portal

Geo-INQUIRE Metadata Schema:

EPOS-DCAT-AP is a dialect of DCAT-AP

EPOS DCAT-AP



TA_FFM-Tohoku

AUTHORS: Jeremy Wong | Zihua Niu | David Schneller | Iris Christadler | Alice-Agnes Gabriel

Edit Experiment | Edit Version | Edit Bulk | Download

DESCRIPTION

Geo-INQUIRE TA Project FFM-Tohoku (C3-TA2-541-11-3, Jeremy Wong, Scripps, UC San Diego). Mesh, input files, and results for five different rupture models for the 2011 Tohoku-Oki Earthquake. For more details, see README.txt. Dynamic rupture simulation ensemble runs generated with SeisSol on Leonardo@CINECA. Related publication: `Dynamic Viability of Finite-Fault Slip Models with Ensemble Kalman Inversion by Wong et al.

Executions | Workflows

exp51

Type to filter files

Filename	Size
J1	8 items
J2	5 items
J3	6 items
J4	8 items
J5	5 items
Mesh	3 items
README.txt	594 bytes

0 selected / 7 total

README preview: README.txt — Markdown

Simulation output for the study `Dynamic Viability of Finite-Fault Slip Models with Ensemble Kalman Inversion` by Wong et al..

Here are a subset of simulation output of the best-fitting rupture models.

J1: Yokota et al., 2011 J2: Minson et al., 2014 J3: Bletery et al., 2014 J4: Melgar and Bock, 2015 J5: Yamazaki et al., 2018

In each model directory, there are model setupfiles in `seissol_param`, and model output files.

Add Version

VERSIONS

Version 1.0.0 — 08/07/25

DOI 10.82554/sdl-51.26

OGC Services

Objects: 149

Data volume: 100.38 GB

FAIR SCORE — LATEST ASSESSMENT

100 /100

F	A	I	R
100	100	100	100

ASSESSED 14/09/2026

How to cite

Experiment version metadata

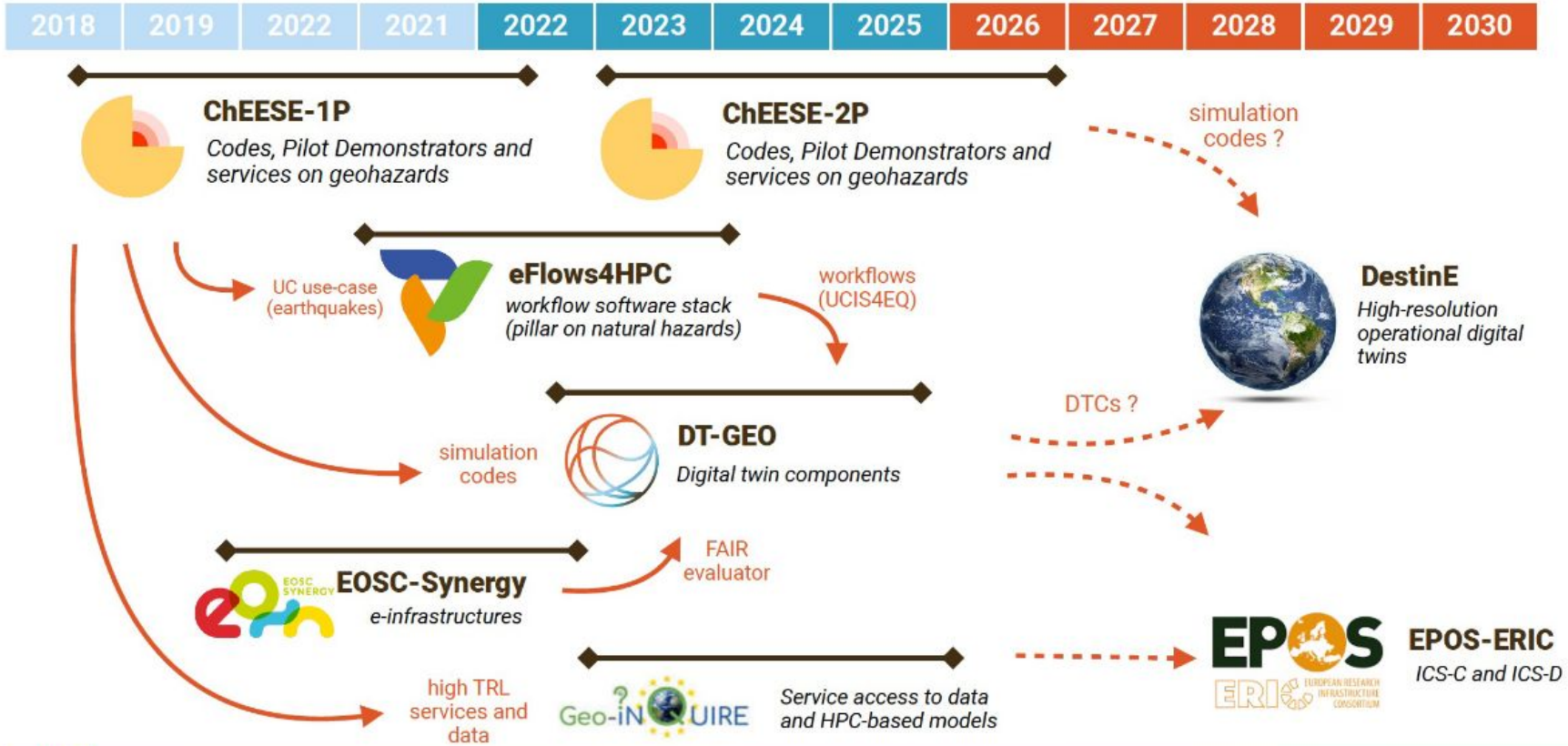
Simulation Details

Dataset Details

Workflow File Details

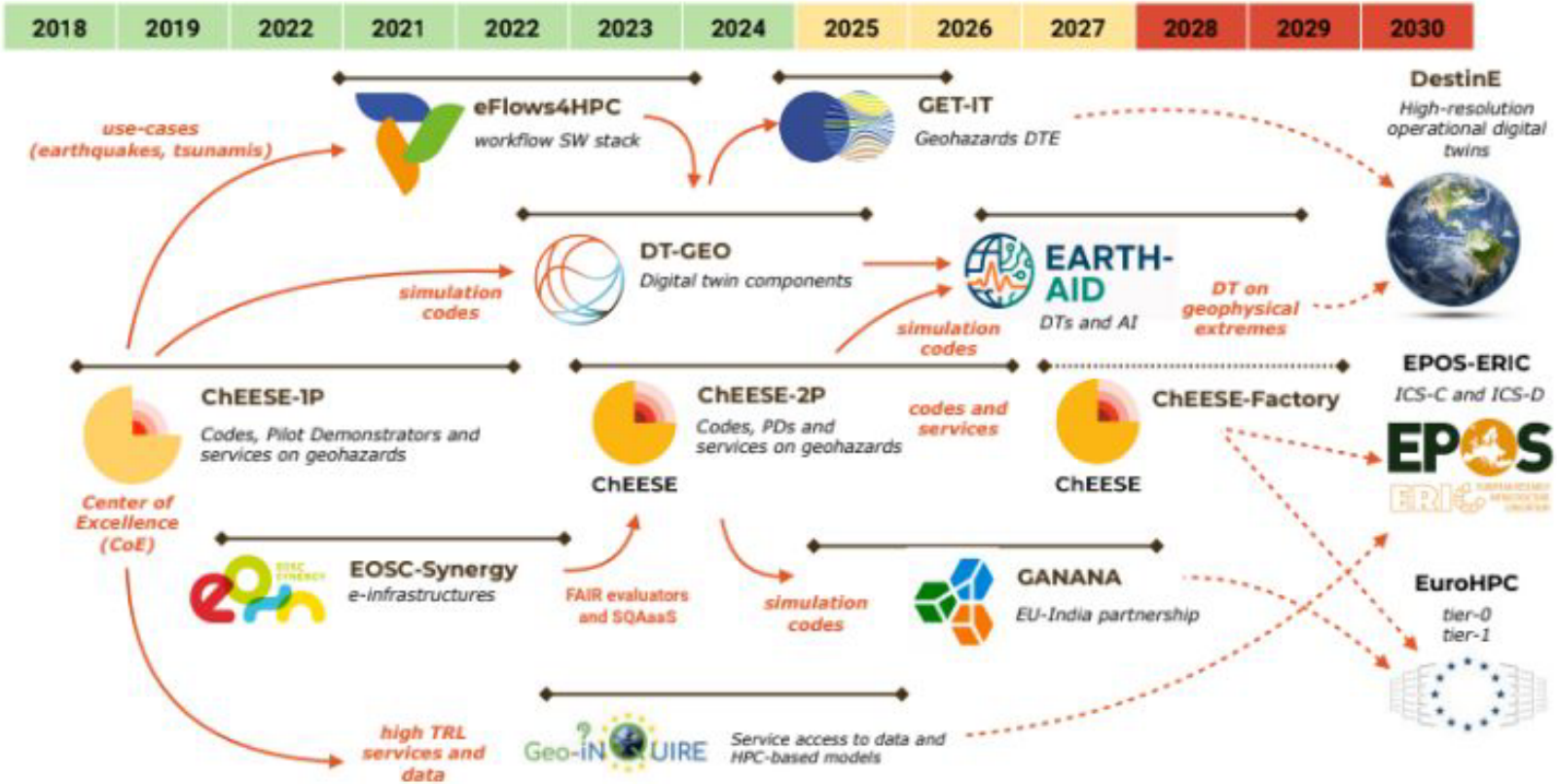
SeisSol & European Projects

Geo-INQUIRE was part of the Destination Earth Roadmap



Roadmap of EU Projects: Arnau Folch (CSIC)

Destination Earth Roadmap (updated)



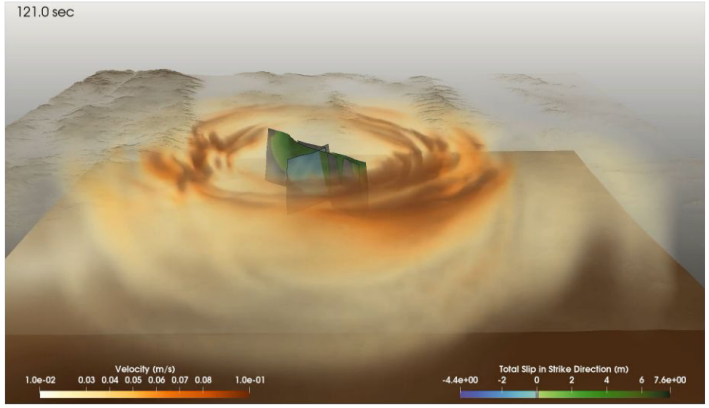
Roadmap of EU Projects: Arnau Folch (CSIC)

The Simulation Data Lake (SDL@CINECA) will play a (key) role in:

- ChEESE-2P (ends Dec'26)
- EarthAID ("follow-on" of DT-Geo, kickoff Oct'26 in Lugano)
- **NFDI4Earth Lighthouse Use Case LMU/TUM** (Oct'26-Dez'27)
- **ChEESE-Factory (aka "ChEESE3")** (to start in Jan'27)

SeisSol in a Nutshell

- **open-source** community code for **physic-based 3D dynamic rupture simulations of Earthquakes**
- uses **Discontinuous Galerkin discretization** and **local time-stepping on unstructured adaptive tetrahedral meshes.**
- Scalable performance at Petascale has been demonstrated **up to several thousands of nodes**
- Joint development by **LMU (Alice Gabriel, "Computational Seismology")** and **TUM (Michael Bader, "Hardware-aware algorithms and software for HPC")**



Propagation of seismic waves (2019 Ridgecrest earthquakes). Visualization of 15 TB of 3D volumetric data on unstructured tetrahedral meshes on Frontera. Taufiqurrahman et al., Nature, 2023
Abrams et al., SC23, Visualization showcase

SeisSol on Github: https://github.com/SeisSol/SeisSol/ doi.org/10.5281/zenodo.4672483

SeisSol Documentation: https://seissol.readthedocs.io/

SeisSol Training: https://github.com/SeisSol/Training
+video introductions online

Based on Slides from Alice-Agnes Gabriel

References:
Breuer et al.,ISC14, Heinecke et al.,SC14,
Breuer et al.,IEEE16, Heinecke et al.,SC16, Rettenberger et al., EASC16
Uphoff & Bader, HPCS'16, Uphoff et al., SC17
Wolf et al., ICCS'20, Uphoff & Bader, TOMS'20
Dorozhinskii & Bader, HPC Asia'21, Gabriel et. al., SC25

From “hero runs” to everyday examples

5 Hz Turkey simulation (larger area):

- required 9h on full (~8200 nodes) of Frontera,
- and a large mesh with 685 millions cell,
- + polynomial order 5 basis functions

“Everyday Example” (Typical model size):

- meshes of ~10M elements,
- 200m spatial sampling of fault geometry,
- 200m finest sampling of topography,
- seismic wavefield up to 2 Hz in fault vicinity
- Comp. costs ~10,000 CPUh

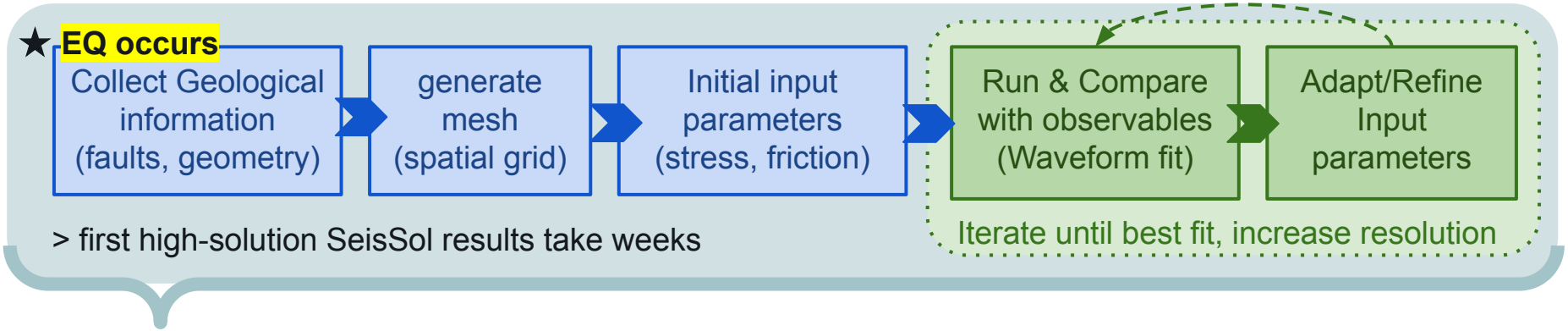
Increases in compute power now allow for **many calibration runs, ”catalog generation”** and even **uncertainty analysis** with UM-Bridge > Increasingly data-rich


Frontera@Univ. Of Texas (TACC)

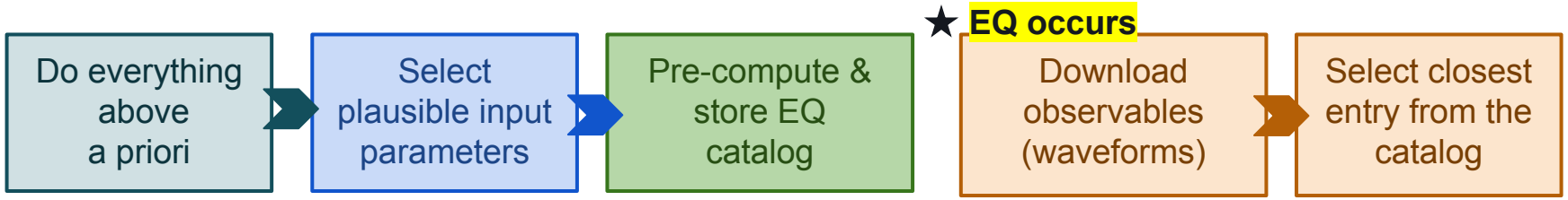

SuperMUC@LRZ

Based on Slides from Alice-Agnes Gabriel

Usual approach takes weeks



Solution: Pre-computed catalog for rapid response



> results within minutes

DT-Geo Demonstrator: Alto-Tiberina Catalog

Observation data & physics

Multi-data constraints from Near-Fault Observatory **TABOO**

-Fault geometry
-Stress orientation
-Frictional properties
-Medium properties

NEAR-FAULT OBSERVATORIES

Access data available within **min to hours** after the eventa

Moderate to **large event** occurrence

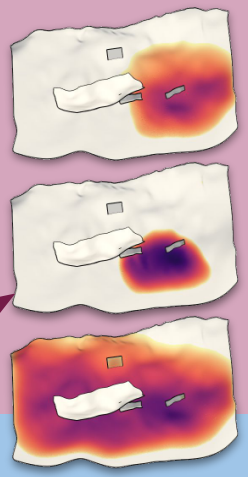
Simulation Workflow

SeisSol

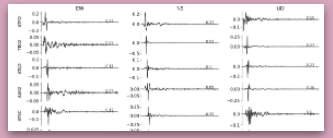
3D dynamic rupture & seismic wave propagation

Varying nucleation location and values of unconstrained parameters

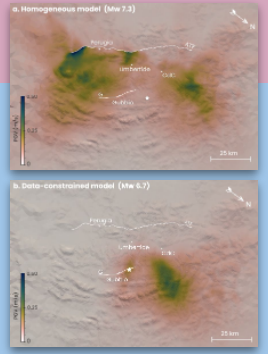
Catalog of **physic-based rupture scenarios…**



… with associated **synthetic data** (ground motion, static displacement, etc)...



… and associated **shake maps**



Search the catalog & output the best fitting scenario(s) (by comparing observations and synthetics)

FAIR Data for follow-on research

© Mathilde Marchandon (LMU)

SeisSol_ParamStudies

Setup parameter studies easy & fast!

4 easy steps:

1. Draft CSV File
2. Turn input files and slurm file in jinja2 templates
3. Add your account
4. Automatically generate input data and batch scripts

Saves you hours in writing/maintaining your own scripts!

```
!LuaMap
  returns: [forced_rupture_time]
  function: |
      function f (x)
        xha = { {{xha_x}},{{xha_y}},{{xha_z}}
```

2. Simply replace variables by {{var_name}}

id	nucleation	d_c	y	mu_s	R	xha_x	xha_y	xha_z
0	1	0.4	0.6	0.35	0.65	295214	4817590	-8238
1	1	0.4	0.6	0.35	0.7	295214	4817590	-8238
2	1	0.4	0.6	0.35	0.75	295214	4817590	-8238
3	1	0.4	0.6	0.4	0.7	295214	4817590	-8238
4	1	0.4	0.6	0.4	0.75	295214	4817590	-8238
5	1						17590	-8238
6	1						17590	-8238
7	1	0.4	0.7	0.35	0.65	295214	4817590	-8238
8	1	0.4	0.7	0.35	0.75	295214	4817590	-8238
9	1	0.4	0.7	0.4	0.7	295214	4817590	-8238

1. CSV File

3. Add your Account

```
if user_name == "IRIS":
    print("Building a slurm script for Iris")
    tp_slurm_account = 'di35ban'
    tp_slurm_email = 'iris.christadler@lmu.de'
    tp_output_file_dir= '/hppfs/scratch/0A/di35ban/AltoTiberina/outputs'
elif user_name == "MATHILDE":
```

4. python3 Scripts/main.py

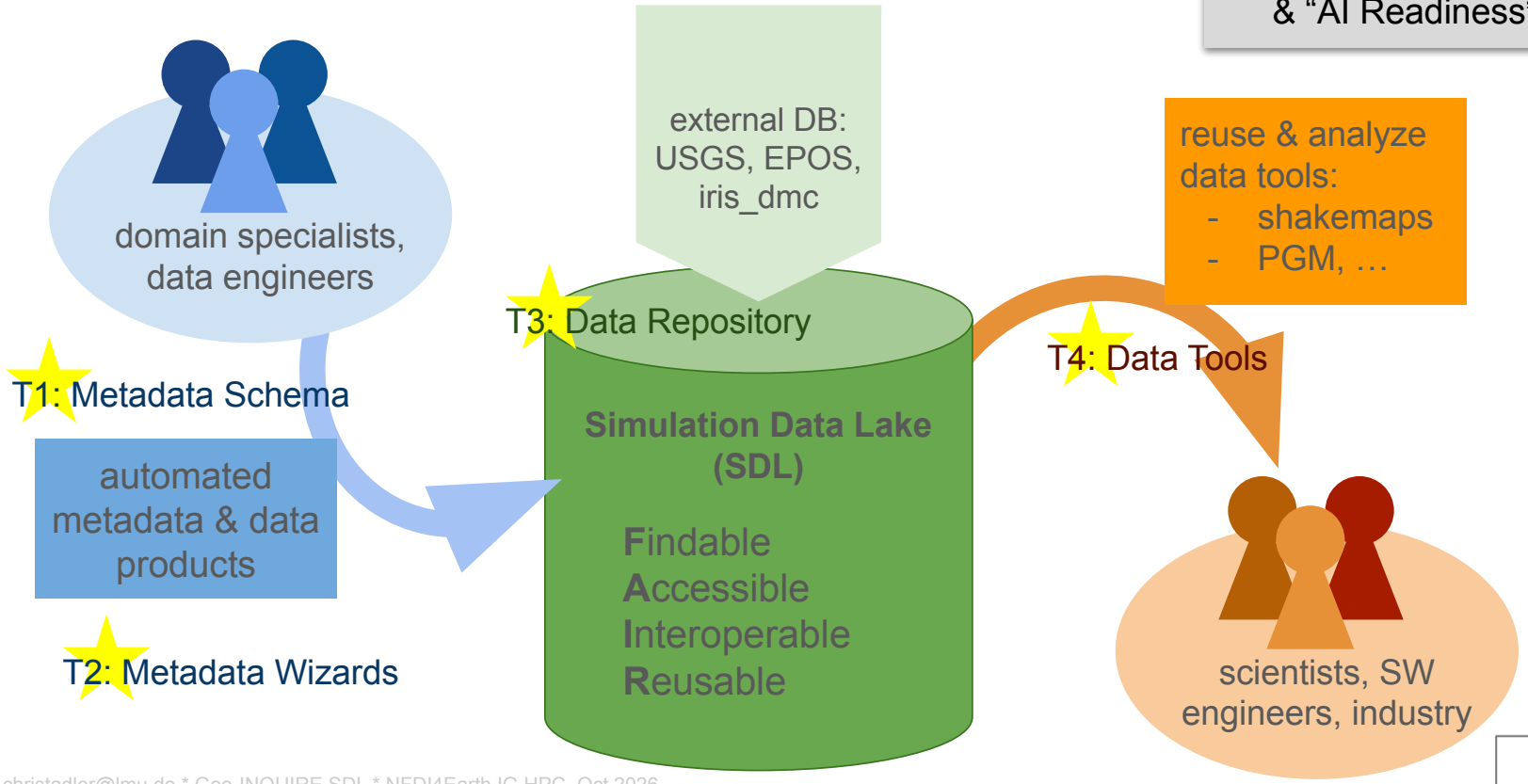
Requires:
Numpy
Pandas
Jinja2

Download now for free(!) from
https://github.com/christadler/SeisSol_ParamStudies

NFDI4Earth LUC: LMU+TUM

“Enabling HPC Earthquake and Tsunami Simulation Data Mining”

“Research Data Management” (RDM) & “AI Readiness”



LRZ Data Science Storage (an SDL alternative?)

- Data Science Storage (DSS)
  + Data Science Archive (DSA) (250 PB of data)
- DOI minting
- Data Cite Metadata Schema
  https://datacite-metadata-schema.readthedocs.io/
- Data also accessible via Globus
  https://www.globus.org/
- Globus Data Transfer
  https://www.globus.org/data-transfer

Link: https://www.lrz.de/presse/ereignisse/2024-04-05-Fair-Data-Portal-dt/

Documentation: https://doku.lrz.de/data-science-storage-10745685.html

LRZ FAIR Data Portal: https://rdm.lab.lrz.de/

www.lrz.de / presse / ereignisse / 2024-04-05-Fair-Data-Portal-dt

Forschungsdaten finden und weiter nutzen



Durch das neue FAIR Data Portal werden Forschungsdaten, die auf den Computersystemen des Leibniz-Rechenzentrums berechnet und gespeichert wurden, auffindbar und weiter nutzbar.

Wetterdaten aus Bayern oder Italien, Simulationen von Erdbeben oder zur Entstehung von Sternen, Modelle von Organen und Gefäßen des Menschen: Die Datenspeicher des Leibniz-Rechenzentrums (LRZ), etwa der Data Science Storage (DSS) sowie das Data Science Archive (DSA) zur längerfristigen Archivierung, können mehr als 250 Petabyte an Daten aufnehmen, die auf den Clustern für High-Performance Computing (HPC) oder den Systemen für Künstliche Intelligenz (KI) erzeugt wurden. „Die meisten Datensätze sind zu groß für die üblichen Forschungsdaten-Repositorien, etwa die von Uni-Bibliotheken", beschreibt Dr. Stephan Hachinger, Leiter des LRZ-Team für Forschungsdatenmanagement, das für das Portal verantwortlich zeichnet, ein grundsätzliches Problem. „Wir versuchen daher, sie direkt von DSS und DSA aus öffentlich zu machen, und im FAIR Data Portal zu präsentieren."



Via this portal, data sets generated at the LRZ - with focus on large HPC datasets - are published according to the FAIR principles.

Feel free to search in the box above!

LRZ FAIR Data Portal: *übersichtlich und intuitiv zu bedienen.*

LRZ Data Science Storage

- Store vast amounts of data
- Access this data from LRZ ecosystem (SuperMUC, Comp. Cloud, AI-Sys, Vis, …)
- Share between arbitrary LRZ users
- Access/Transfer this data world wide via a high performance, WAN optimized transfer protocol,
- Share with arbitrary users around the globe
- Additionally: **Data Science Archive (DSA)**

- Special LRZ "**data projects"** (organizational envelope of the data, grants from one or more data pools)
- Data "**curators**" (project users, typically PI and one proxy, "manage, own and take responsibility for the data"
- "**Containers**" (created by data curators) manage through DSSWeb Portal:
  - Access rights & data limits (in summary & per user)
  - export container to Cloud or VMWare
  - if data in the container should be protected by regular tape backups or even archives
- Containers can be shared via **Globus Online**

Infrastructure

- consists of multiple storage building blocks, integrated to a single virtual storage system, using IBM Spectrum Scale SW (up to PBs)
- storage building blocks are connected to a dedicated high performance interconnect (high speed connections to all of LRZ's compute systems)
- **DSA** additionally has two tape libraries in separate data centres attached that hold the frozen data.

Geo-INQUIRE Transnational Access (TA) Datasets

LMU TA Projects:

Overview of TA reports https://www.geo-inquire.eu/transnational-access/project-reports

SeisSol/ExaHyPE TA:

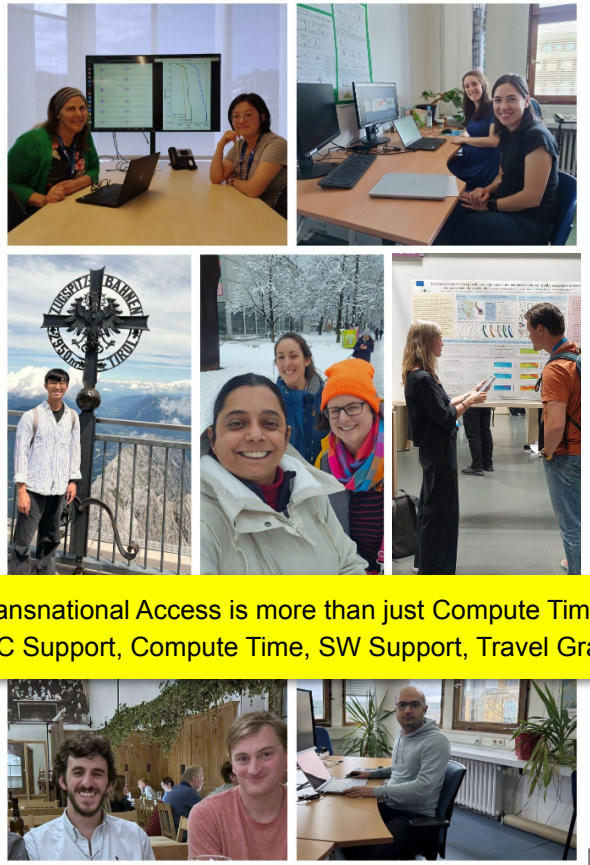
- **Haiti-drm** *(Zoe Yin, UC San Diego, US)*
  Dynamic rupture model of the 2021 M7.2 Haiti Earthquake
- **CapCosHPC** *(Elizabeth Madden, San José State University, US)*
  Capturing co-seismic stress changes with HPC enabled dynamic rupture models of megathrust earthquakes
- **RADIATE** *(Francesco Mosconi, La Sapienza, Uni Rome, Italy)*
  Large ensemble earthquake simulations (Bedretto)

MP-PSHA TA:

- **FFM-Tohoku** *(Jeremy Wong, UC San Diego, US)*
  validating finite-fault models of the 2011 Tohoku-Oki earthquake with ensembles of dynamic rupture simulations
- **PhyDRMT** *(Neha, National Taiwan University, Taiwan)*
  Physics-based Dynamic Rupture Modeling for Taiwan

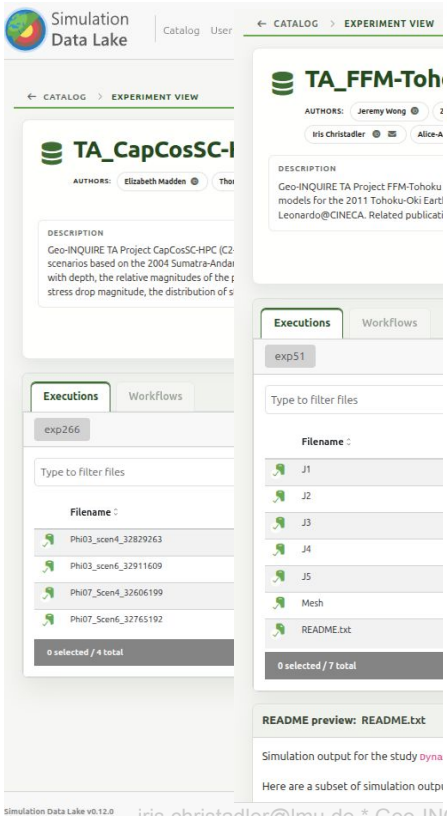
CyberShake-PSHA TA:

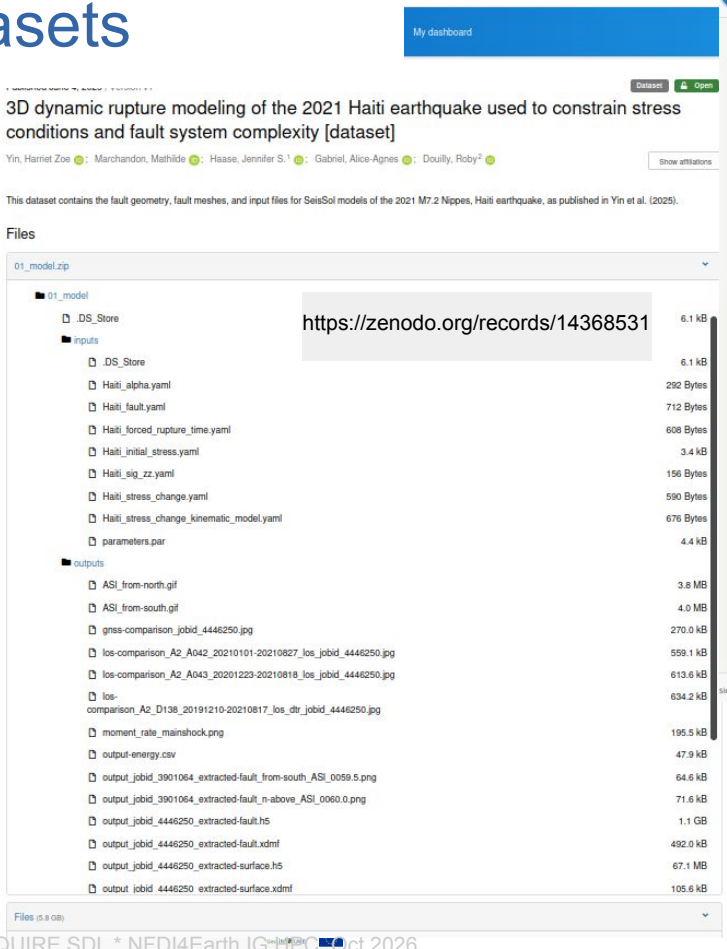
- **CyberSISZ** *(Hamed Davari, Univ. of Iceland, Iceland)*
  CyberShake finite fault rupture modeling for improved near-fault ground motion simulation of strong earthquakes in Southwest Iceland
- **SHARE** *(Duo Li, GNS Science, New Zealand)*
  Seismic HAzards from multi-fault Ruptures of crustal and subduction Earthquakes

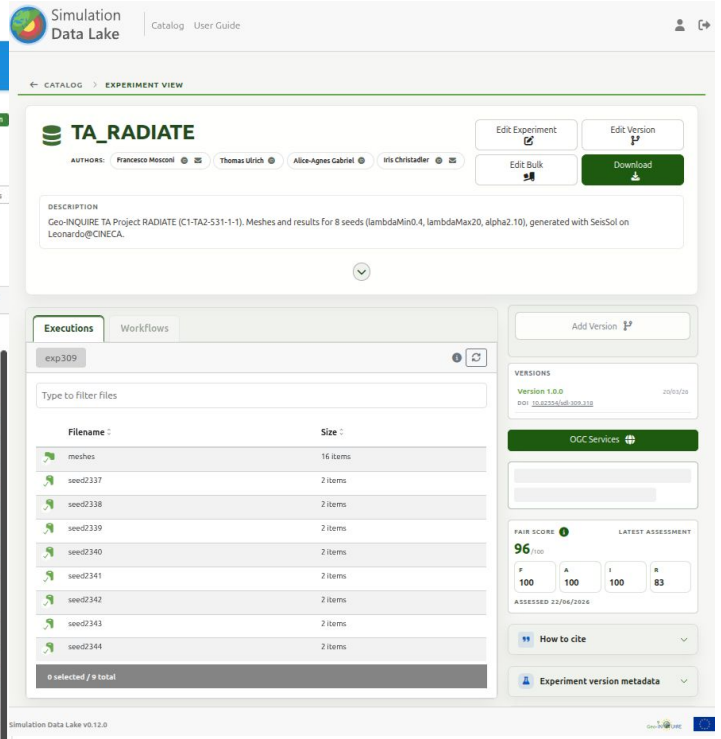


Transnational Access is more than just Compute Time:
HPC Support, Compute Time, SW Support, Travel Grant

SeisSol TA Datasets







Each Researcher found its own way of packing data 😆

SeisSol TA Datasets: Data Layout Summary

Mesh
Input
Output
Job Scripts
Simul.Runs

Haiti-drm:

- Inputs
  - parameters.par, *.yaml
- Outputs
  - *.xdmf, *.h5, *.csv (SeisSol output)
  - *.png (data products)

CapCosHPC: (Mesh in a different Repo)

- Phi0X_scenX_jobid (4 simulation runs)
  - Parameters.par, *.yaml
  - slurm.sh, slurm.out, slurm.err
  - Output_scenX_normal
    - *.xdmf, *.h5, *.csv

RADIATE:

- Meshes:
  - .xdmf, .h5 (one mesh for each simulation)
- seedX
  - Output
    - *.xdmf, *.h5, *.csv, *.receiver.dat

FFM-Tohoku:

- Jx
  - Seissol_param (Input):
    - *.par, *.yaml
    - *.sh
  - *.xdmf, *.h5 (output)
- Mesh:
  - *.xdmf, *.h5 (one mesh for each simulation)
- Readme.txt

PhyDRMT:

- Input:
  - Parameters.par, *.yaml, receivers.dat
  - sbatch.sh, *.out (stderr+stdout), select_gpu
- Mesh:
  - .xdmf, .h5 (one)
- Readme.md

SeisSol/HPC Data Layout https://github.com/christadler/SeisSol_ParamStudies

$WORK/$HOME:

- **Executable:**
- **Asagi-file:**
- **Mesh:**
  *.xdmf, *.h5
- **Input:**
  Job Farming Batch Scripts
  Batch Output
  - Input01:
    Parameters.par, *.yaml, *.dat
    (Batch Script)
  - Input02: …
  - Input03: …

$SCRATCH:
(high watermark or other regular deletion policy)

- **Output**:
  - Output01_jobID: (raw output & data products)
    *xdmf, *.h5, *.csv
  - Output02_jobID:
  - Output03_jobID:

Freenas@LMU:
(data copy for visualization and to save data)

- **Output: (data products & selected timesteps)**
  - Output01_jobID
  - Output02_jobID …
- **Mesh:**
  - Mesh01
  - Mesh02 …
- **Input:**
  - sbatch.sh, jobid.out, jobid.err
  - Input01
  - Input02 …

Which output or data products need to be stored depends on your scientific research question

SDL_Metadata_Generator https://github.com/fynngraf/SDL_Metadata_Generator

Goal:

- Easy script for all SeisSol users to encourage them to use the SDL

Starting Point:

- SDL Metadata is (currently) mainly **file formats, types and path**
  > Can be generated by analyzing the directory

Idea:

- Analyze data and provide easy scripts to gather data (tbd)
  Input, Mesh, Output > README.md
- Get some specific information from user through README.md
- Use a script to build metadata.json
- Add SeisSol metadata (tbd):
  - Information on the environment and run
    (code version, system, FLOP/s… from SLURM file/outp
  - SeisSol Input Metadata
    (from parameters.par and additional .yaml files)
  - SeisSol Output Metadata
    (from output files, data products)

The SDL Metadata Wizard:

- Is intended for smaller datasets
- It needs many information
  but can't be scripted
- Non-deterministic (running it twice on the same input data set yields different metadata.jsons)

Could be used/adapted by many different codes

```
cp templates/example.README.md  \
    $WORK/sdl_data_new/README.md

vi $WORK/sdl_data_new/README.md

sdl_metadata_generator  --path=$WORK/sdl_data_new
```

3(!) easy steps:

1. Draft README.md (yaml)
2. Generate metadata.json
3. Upload metadata to SDL

SDL Metadata Generator

Reproducible metadata in minutes!

If the SDL Wizard is just not yours ;)

```yaml
name: "NewExperiment"
description: "One or two sentences describing what
author: ["36", "82"]
version: "1.0.0"
keywords: ["seissol", "example"]

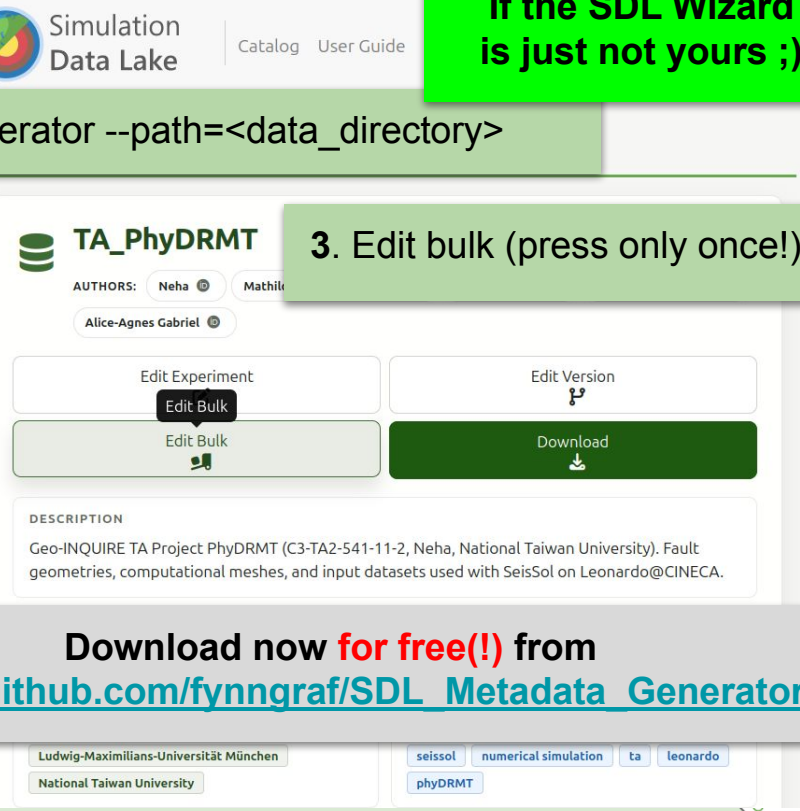
folders:
  # Description for a top-level run/model folder (used as that folder's
  # own description), or for a structural folder that repeats identically
  # under every top-level folder.
  run1: "Rupture model run1"
  run2: "Rupture model run2"
  seissol_param: "{folder}: Input files"

extensions:
  # Description by file extension (include the leading dot).
  .sh:
    description: "{folder}: Slurm batch scheduling script"
    priority: true
  .par:
    description: "{folder}: SeisSol main parameter file"
    priority: true
  .yaml:
    description: "{folder}: Additional SeisSol parameter files"
    priority: true
  .xdmf:
    description: "{folder}: SeisSol xdmf info to use bin data from .h5 files"
```

1. README.md

2. sdl_metadata_generator --path=<data_directory>

3. Edit bulk (press only once!)



Requires:
pyyaml
Jinja2

Download now for free(!) from
https://github.com/fynngraf/SDL_Metadata_Generator

SeisSol Metadata: First Proposal -> Reproducibility

Environment

ORDER, EQUATIONS
PRECISION
Github Commit No.
Platform (slurm script, .bashrc or env/modules?): LUMI, Leonardo, SuperMUC, …

Executable

EESSI European Environment for Scientific Software Installations (tbd)
Github Commit (meanwhile)

Parameters

Nucleation point
Mu_d, mu_s
D_c, R
SHmax
FL (0: none, 16:LSW, 103: RS with strong velocity weakening)
Plasticity?
cICType?

Dump all parameters

Mesh

Results/Data Products?

Ground Motion Parameters?
Shake Maps?
What else?

Hands on: SDL

SDL Demo

Web Interface:

- *Available (published) data*
- *Much more data if you have a login*
- *Look at FFM_Tohoku:*
  - *Project metadata*
  - *Readme.txt*
  - *General structure*
- *Hands-on: Generate a new Repo*

Command Line Interface (on Leonardo):

- *Log in, update*
- *Experiment list*
- *Download data (~500MB)*
- *Upload data to newly generated repo*

Generate Structure from metadata.json using SDL_Tools:

- *Download (RO-CRATE) metadata*
- *compare different metadata.json*
- *Generate empty structure from metadata.json*

Generate Metadata with SDL_Metadata_Generator:

- *Copy and adapt README.md*
- *Generate metadata.json*
- *Edit bulk>metadata.json (takes a while)*

SDL Quick Start / Cheat Sheet

```
# Leonardo
source ~/venv/sdl2/bin/activate

# install SDL toolset
sdlctl --version
pip install sdlctl -U --index-url
https://gitlab.hpc.cineca.it/api/v4/projects/1647/p
ackages/pypi/simple
sdlctl --version
sdlctl conf setenv prod
sdlctl conf getenv
sdlctl user login
sdlctl experiment list

### WIZARD (for metadata.json)
sdlctl wizard sdl_100runs/
sdlctl experiment bulk -file metadata.json
-format json
sdlctl -y file bulk -id 309 -v latest -lp
TA_RADIATE
```

```
# Upload folder one at a time
sdlctl file bulk -id 86 -lp /sdl_100runs/0001 -dp 0001

# Uploading one file to an arbitrary location
sdlctl file upload -id 661 -v 1.0.0 -lp
/leonardo/pub/userexternal/emadden0/topo4_splays_fi
x9-14...suma -dp Meshes

### DOWNLOADING FOLDER
sdlctl -y experiment dump -id 261 -v 1.0.0 -e
data -ff inputTW0004

### DOWNLOAD FULL EXPERIMENT
sdlctl -y experiment dump -id 48 -e data # only
data
```

Take-home messages

- SDL: attractive solution for scientists (but scientific metadata is rather sparse)
- LRZ Data Science Storage is an alternative solution

- While SeisSol can be used for hero runs,
  more and more computing time is spent in “everyday” scenarios
- Increases in machine size and capability now enable bigger parameter studies
  > Leading to an significant increase in data
- Research data management for SeisSol is becoming increasingly important

- NFDI4Earth LUC LMU+TUM will (most probably) build on SDL
- Extended/Scientific Metadata Schema is necessary to make useful use of the data in the SDL
- Workflows for automatic storing and retrieving are planned

Thank you for your attention!

